Anthony Ngon

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TECHNICAL SKILLS

Languages: TypeScript, Python, Java, Scala, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Three.js, Tailwind CSS, Node.js, PySpark, Apache Spark, Kafka, Pandas

Databases: PostgreSQL, MySQL, Oracle, Snowflake

Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, CloudWatch), Azure, Google Cloud Platform, Kubernetes, Docker, Terraform

DevOps & Testing: Jenkins, Git, CI/CD, Vite, Vitest, Selenium, JUnit

Certifications: DP-900: Microsoft Azure Data Fundamentals

EXPERIENCE

Software Engineer, Data Platform **Jul 2022 – Jun 2025**

Optum (UnitedHealth Group) – Norwalk, CA

- Designed and deployed automated ETL pipelines in PySpark, Python, and SQL processing 50K+ daily healthcare records, reducing manual data handling by 40% and cutting end-to-end latency from hours to under 15 minutes
- Built and maintained RESTful microservices on Azure serving 3 downstream applications with Medicare claims and member enrollment data, enabling marketing and analytics teams to access validated datasets in real time with 99.9% uptime
- Architected CI/CD pipelines with Jenkins, Docker, and Terraform, automating test and deployment workflows that reduced release cycles from weekly to daily and cut production incidents by 30%
- Engineered a shared Python utility library adopted by 4 engineering teams (~20 developers), standardizing data validation and transformation patterns and reducing duplicated code by an estimated 25%
- Orchestrated 10+ data workflows in Databricks and Azure Data Factory spanning both batch and real-time streaming pipelines, processing Medicare claims and enrollment data on Kubernetes to meet daily SLA windows during peak enrollment periods
- Stood up AWS infrastructure using EC2, S3, and Lambda to support cross-cloud data transfer jobs, migrating legacy batch exports to event-driven pipelines that cut delivery time by 35% and eliminated manual file handoffs between teams
- Monitored pipeline health and set up CloudWatch alarms for anomaly detection across AWS-hosted jobs, reducing mean time to detection on data issues from hours to under 10 minutes
- Mentored 3 junior engineers through code reviews, pair programming, and writing onboarding documentation that became the team's standard reference for new hires

PROJECTS

CSUF Campus Navigator | Next.js, TypeScript, React, Tailwind CSS, Leaflet

- Built a full-stack campus navigation app featuring an interactive Leaflet map, searchable directories, and real-time event discovery for 63 buildings across Cal State Fullerton
- Engineered a complete ground-up rebuild from vanilla HTML/JS to a modern React architecture, reducing marker rendering code from 360 lines to ~30 lines through data-driven component design
- Implemented client-side route calculation using the haversine formula, eliminating external API dependencies and achieving a zero-API-key architecture

Streets Alive – 3D City Simulation Engine | Three.js, React, TypeScript, Vite

- Built a real-time 3D city rendering engine that decodes OpenStreetMap vector tiles and renders extruded buildings, multi-lane roads, and highway elevation tiers for the Los Angeles area
- Designed a life simulation layer with population modeling, need-based decision making, and A*-routed car traffic animating across a dynamic road network
- Optimized rendering performance using unified GPU buffers, LRU + IndexedDB geometry caching, a parallel Web Worker pool for tile decoding, and async yielding to maintain 60fps

EDUCATION

Bachelor of Science in Computer Science May 2022

California State University, Fullerton